Kunjal Purohit

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Education

University of California, Santa Cruz — Santa Cruz, CA

Bachelor of Science in Computer Science and Bachelor of Science in Computer Engineering — *Expected June 2028*

- **GPA: 3.95/4.0**, Dean's Honors List (3x)
- **Relevant Coursework:** Data Structures & Algorithms, Computer Systems & C, Assembly Language, Discrete Mathematics, Probability & Statistics, Linear Algebra, Multivariable Calculus, Differential Equations

Skills

Languages: Python, TypeScript, JavaScript, C/C++, Java, Swift, SQL (PostgreSQL), HTML/CSS, Assembly
Frameworks & Libraries: Next.js, React, Node.js, FastAPI, Capacitor, Convex, Playwright, Tailwind CSS, OpenCV
Tools & Platforms: Supabase, Neo4j, Clerk, Plaid, Vercel, Codemagic, Unix/Linux
Embedded Systems: STM32, ESP32, CAN Bus, OneWire, Sensor Integration, Signal Filtering
AI: Coding Agents, LLM Integration, Prompt Engineering

Experience

Formula Slug (FSAE) — October 2025 – Present

Embedded Software Engineer, Electrical Branch — *Santa Cruz, CA*

- Own STM32 C/C++ firmware for real-time vehicle data acquisition, from sensor input to CAN bus telemetry
- Scoped, built, and validated tray-temperature monitoring across 4 DS18B20 OneWire sensors, triggering an **automatic shutdown** on safety-threshold faults
- Replaced a discontinuous piecewise torque-mapping function with a continuously differentiable cubic-polynomial curve, now the **default accelerator mapping** across the car's VCU firmware
- Redesigned the CAN protocol for a new battery and chip architecture, **adopted team-wide** for on-car telemetry

Association for Computing Machinery (ACM) — August 2026 – Present

Web Development Lead — *Santa Cruz, CA*

- Led 4 developers in redesigning the club website, shipping 5 features before fall recruitment
- Built member login with session handling and a mobile-responsive layout, deployed on Vercel
- Redesigned the events page and navigation to showcase 15+ events

Projects

UCSC Dining Tracker | App Store | *Next.js, TypeScript, Supabase, Python, Swift* — June 2026 – Present

- Shipped a web and iOS macro tracker (App Store) reaching **70+ users**, featuring daily target rings, custom macro goals, and meal-period food suggestions
- Built Python/Playwright scraping pipelines executing daily via GitHub Actions with a **100% success rate across 470+ runs**, indexing **1,699 campus food items** and **233K+ menu records** across **14 dining halls**
- Built a Supabase PostgreSQL database with **90+ days of menu history** and per-user Row-Level Security (RLS)
- Automated iOS builds, code-signing, and App Store deployments using a Codemagic CI/CD pipeline

Dowser (Oil Well Detection) | Live App | *Next.js, Convex, Capacitor, ESP32, YOLO* — October 2026

- Built in 42 hours at Cal Hacks 13.0 with a team of 4: detects well symbols on old USGS topo maps, converts them to coordinates, and checks them against **158,000 California state well records** to flag **candidate** wells
- Built the Capacitor iOS field app and Convex backend, streaming ESP32 readings live to a map with auto-confirm logic (8 readings $\geq$ 1000 within 10 s), and set up the ESP32 hardware

Heirloom (AI Multi-Generational Wealth Platform) | *Next.js, FastAPI, Neo4j, Plaid, Daytona* — October 2026

- **Won 1st place in the Daytona track** (20 teams) at SF Tech Week by building an AI wealth platform in 12 hours, integrating AI agents, a Neo4j knowledge graph, and Plaid, and iterating on finance professionals' input
- Designed AI agents that flag conflicts between parents' financial decisions and heirs' goals for wealth advisors
- Architected firm-level data isolation by mapping Clerk JWT claims to PostgreSQL Row-Level Security (RLS) policies, enforcing dynamic, multi-tenant security across all database operations

Autonomous Vehicle (UCSD COSMOS) | *Python, OpenCV, ROS, LiDAR, GNSS, Jetson Nano* — July 2024

- On a 5-person team, wrote Python drivers for a LiDAR and GNSS receiver, fused both into one time-aligned log, and generated synthetic LiDAR logs from a kinematic bicycle model, enabling **3 fully autonomous laps**
- Built occupancy-grid mapping and localization, validated in simulation and on real indoor/outdoor courses